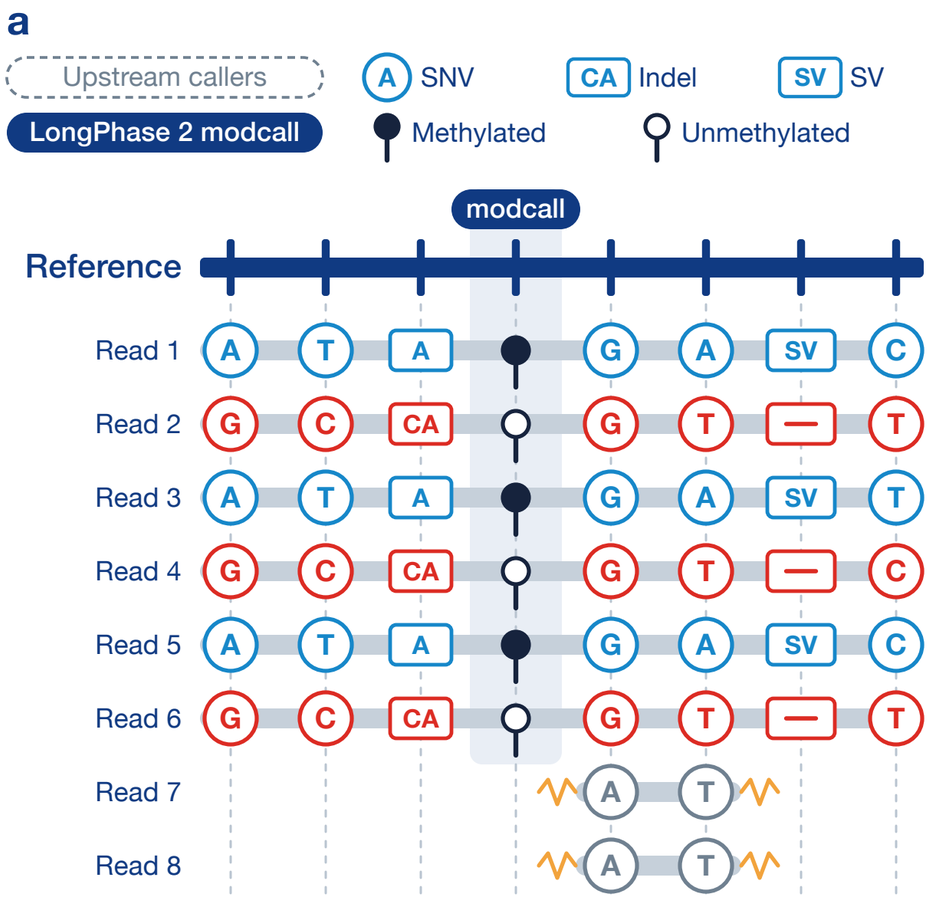
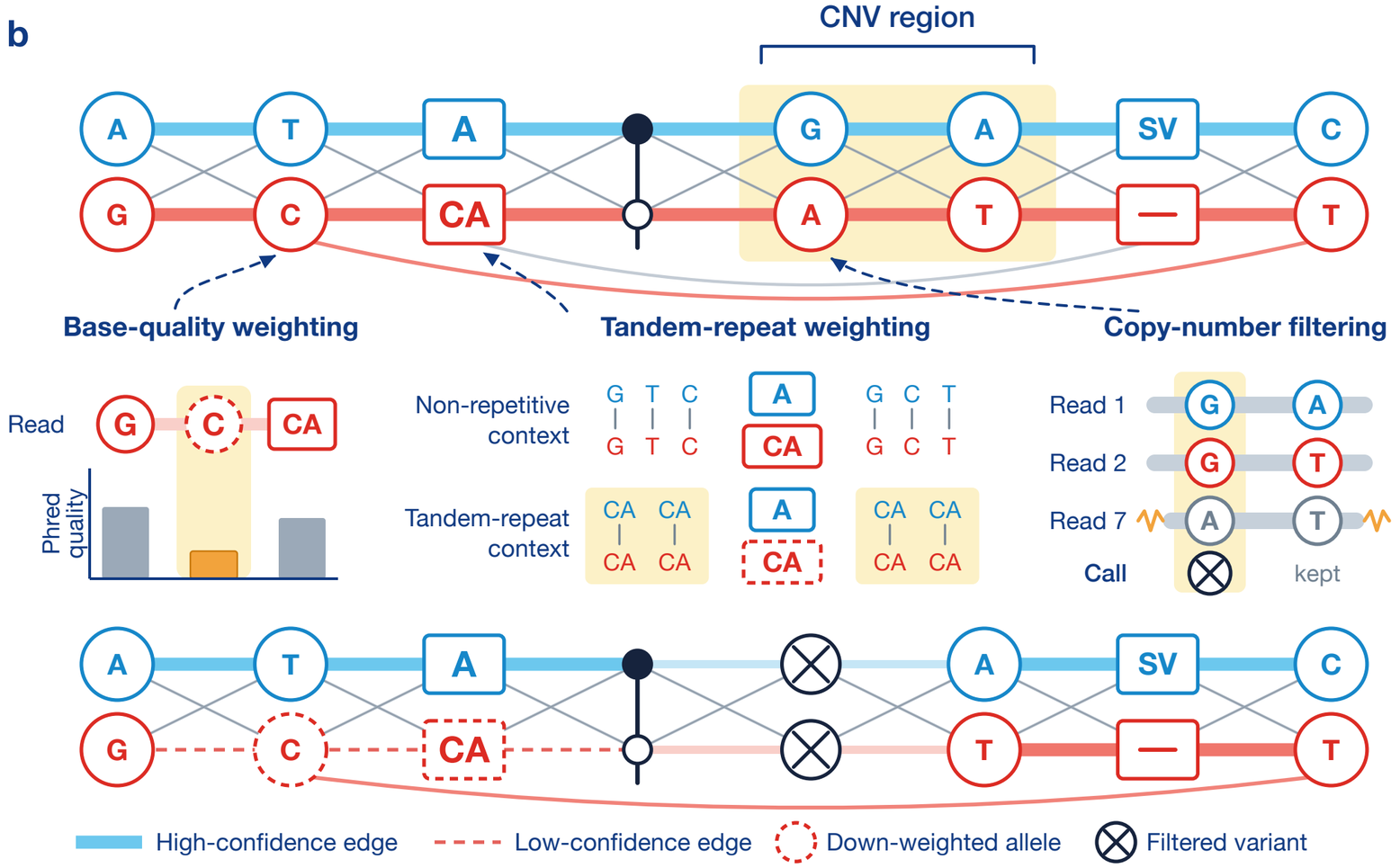
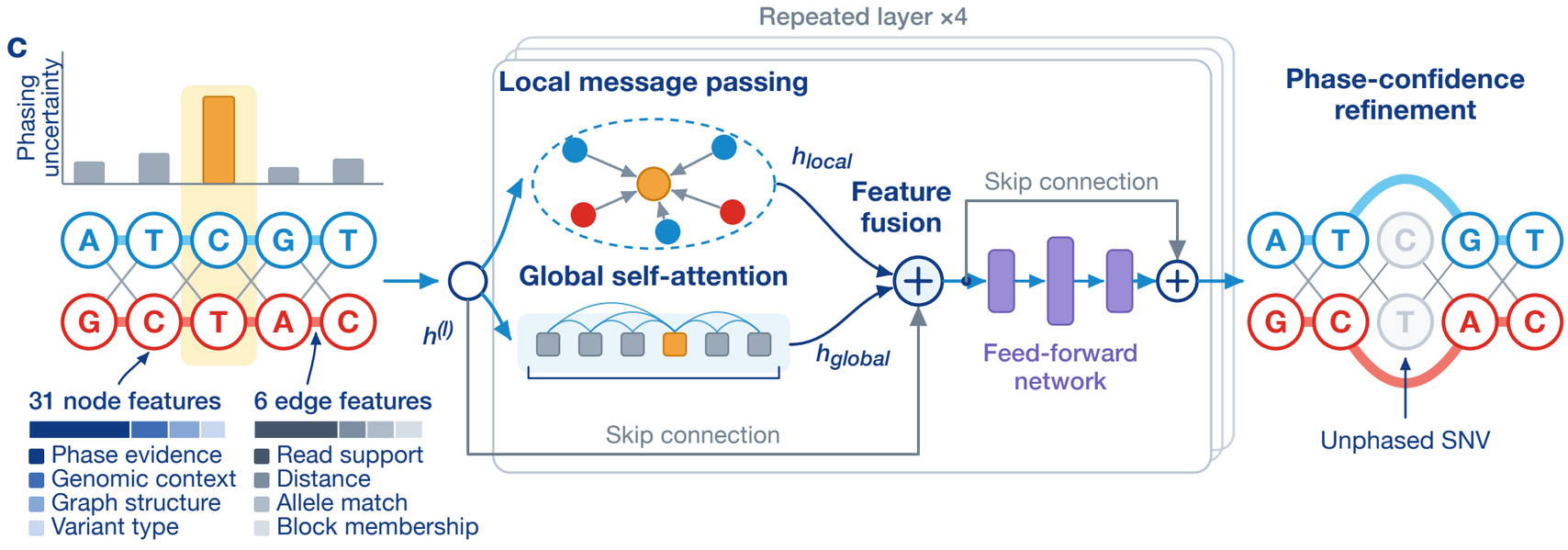
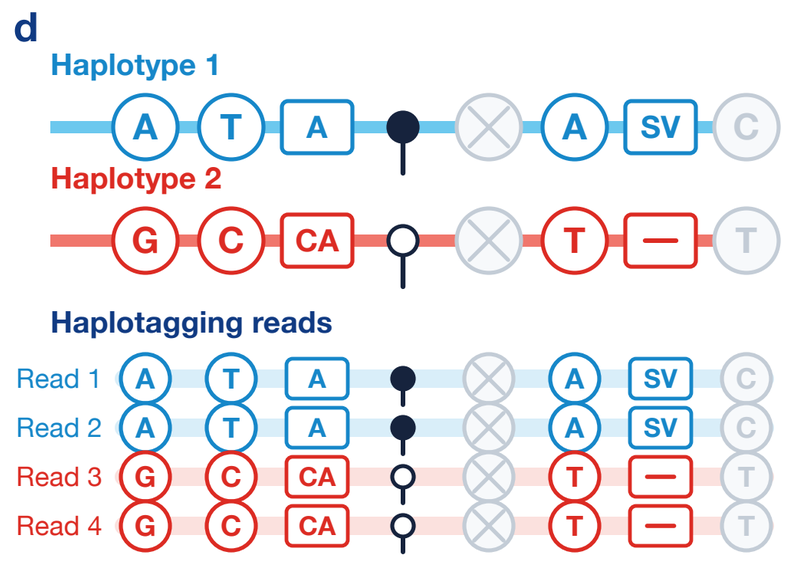

a
Upstream callers
LongPhase 2 modcall
SNV
Indel
SV
Methylated
Unmethylated
modcall
Reference
Read 1
Read 2
Read 3
Read 4
Read 5
Read 6
Read 7
Read 8


b
CNV region
Base-quality weighting
Tandem-repeat weighting
Copy-number filtering
Read
Phred quality
Non-repetitive context
Tandem-repeat context
Read 1
Read 2
Read 7
Call
kept
High-confidence edge
Low-confidence edge
Down-weighted allele
Filtered variant


c
Repeated layer ×4
Phasing uncertainty
31 node features
6 edge features
Phase evidence
Genomic context
Graph structure
Variant type
Read support
Distance
Allele match
Block membership
Local message passing
Global self-attention
Feature fusion
Skip connection
Feed-forward network
Phase-confidence refinement
Unphased SNV


d
Haplotype 1
Haplotype 2
Haplotagging reads
Read 1
Read 2
Read 3
Read 4